

Lab 4: Parallel RLC Circuit

A. Objectives

- Investigate peak voltage, current and phase relationships between the circuit components of parallel RC, RL, and RLC circuits.
- Understand and prove Kirchhoff's Current Law for AC circuits.
- Understand the technique to measure current using a current sense resistor.

B. Background

B.1. Voltage and Current in a parallel AC circuit:

In a parallel RLC circuit, the source voltage V_s is common to all the components and the individual branch currents through each element is different. From our previous experiments we know that for resistors voltage is always in phase with the current, for capacitor voltage lags the current by 90 degrees and for inductors the voltage leads the current by 90 degrees. Because each element has a unique phase response between +90 and -90 degrees, a parallel combination of R, L & C will yield a complex impedance with a phase angle between +90 and -90 degrees. Due to this phase response, Kirchhoff's Current Law must be computed using vector (phasor) sums rather than simply relying on magnitude.

B.2. Measuring current using an Oscilloscope:

We can indirectly determine the magnitude and phase of the current in any branch of a circuit by using a small sense resistor. The phase angle of the current in the branch will be the same as the phase angle of the voltage across the sense resistor.

In the circuit in **Figure B.2.1**, the source current can be measured by adding a resistor ' r_1 ' at **Box 1**. The resistor chosen must be much smaller than the total circuit impedance so as to have minimal impact on the total current. The phase difference between V_s and V_{r1} will be the phase angle of the circuit current.

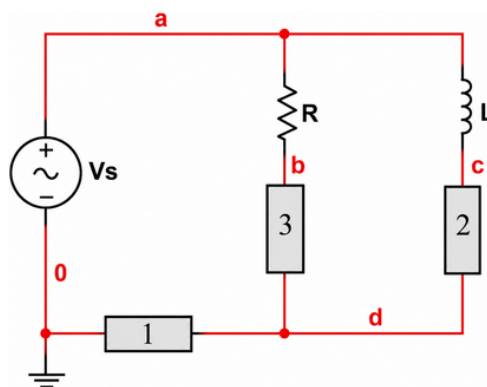


Fig.B.2.1: Sense Resistor Placement

Similarly, to find the current through the inductor L, we can add a sense resistor ' r_2 ' at **Box 2**. The magnitude of the current through L can be determined using Ohm's Law and the phase can be obtained from the phase difference between V_s and V_{r2} .

Experiment 2: Parallel RLC circuits

A.2 Objective

- To analyze the relationship between the current and phase of reactive elements and the source in parallel RC, RL and RLC circuits.

B.2 Theory

Please refer to part B.1 and B.2 in the Background section.

C.2 Apparatus

Components	Instruments
- Resistors: 1×10kΩ, 1×1kΩ - Capacitors: 1×10nF - Inductors: 1× [6 – 14] H	1× Trainer Board 1× Audio Generator 1× Dual Channel Oscilloscope - Connecting wires and probes

D.2 Procedure

D.2.1 Measuring the practical value of circuit components and calculating the impedance

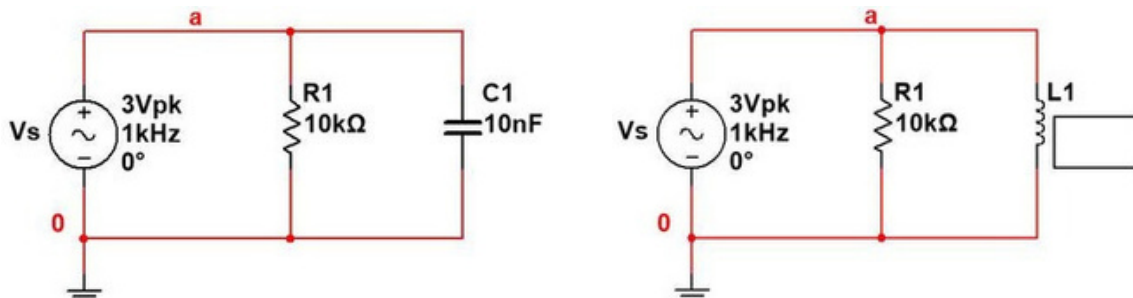
- Measure the practical value of the resistor (R) using DMM and note down the value in Tables 2.1, 2.3 and 2.5. Use the measured values in all your calculations.
- Measure the practical value of the capacitor (C) using an LCR meter and note down the values in Tables 2.1, and 2.5. Do the same for the inductor (L) and note down the values in Tables 2.3 and 2.5 and also in Figures D.2.2 and D.2.3. If an LCR meter is not available, simply note down the nominal values.
- Calculate the Reactance (**X**), magnitude of total Impedance (**|Z|**) and phase of total Impedance (**Z∠θ°**) in Tables 2.1, 2.3 and 2.5 and note down the values in the respective tables.

D.2.2 Setting Up the Oscilloscope

- Calibrate both Channel 1 and Channel 2.
- Set the time/division to 0.2ms.
- Set the voltage/division to 1V for both the channels.

D.2.3 Constructing Circuit 1 (Parallel RC)

- Construct the circuit shown in **Fig.D.2.1** on the bread board. Use minimal wires.
- Set 1 kHz Frequency and 3V peak Amplitude (6V peak to peak) in the Audio Generator.
- Connect Channel 1 of the oscilloscope across the source V_s (positive red port to node 'a' and negative black port to node '0' i.e. ground).
- Observe the generated signal on the oscilloscope screen and fine tune the frequency and amplitude of the input signal generated from the audio generator to match the nominal values. Always set the amplitude after setting the frequency because changing the frequency of a non-ideal source might alter the amplitude.



D.2.4 Calculating the currents through the different circuit components

1. Calculate the magnitude and phase of I_S (I_{source}), I_C and I_R and note down the values in **Table 2.2**.

D.2.5 Measuring the peak current through the source and the time delay with the oscilloscope

1. Use the information in section B.2 to connect the 1k Ω sense resistor (r) at suitable point in order to measure the source current, I_S .

Fig.D.2.1: Parallel RC circuit**Fig.D.2.2: Parallel RL circuit**

2. Connect and use Channel 2 to view the voltage across the sense resistor.
3. Measure the peak voltage across the sense resistor and calculate the peak current using Ohm's Law. Note down both in **Table 2.2**.
4. Change the display mode of the oscilloscope so that both Channels 1 and 2 are visible on the screen.
5. Measure the time between a peak of the source waveshape (V_S – Channel 1) and the next peak of the voltage across the sense resistor (V_{sense} – Channel 2). Note down the time (Delay) in **Table 2.2**.

D.2.6 Measuring peak voltage and time delay across the resistor and capacitor with the oscilloscope

1. Now, move the sense resistor to the branch containing R1 and use the same method you used in the previous step (D.2.5) to measure the peak current in the branch and the time (Delay) between V_S (Channel 1) and V_{sense} (Channel 2). Note down the values in **Table 2.2**.
2. Next, move the sense resistor to the branch containing C1 and use it to determine and record the peak current in that branch and the delay.

D.2.7 Constructing Circuit 2 (Parallel RL)

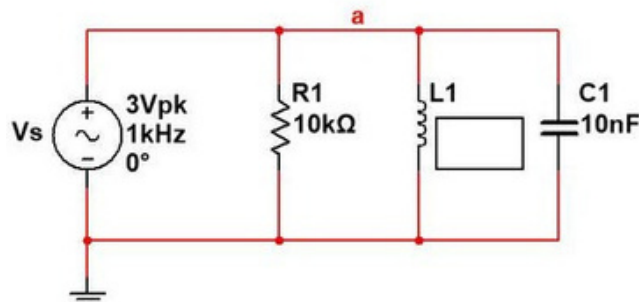
1. Construct the circuit shown in **Fig.D.2.2** on the bread board. To do so, simply replace the capacitor in the previous circuit with one of the provided inductors.
2. Set 1 kHz Frequency and 3V peak Amplitude (6V peak to peak) in the Audio Generator.

D.2.8 Calculating the voltage across the circuit components

1. Calculate the magnitude and phase of I_S , I_L and I_R and note down the values in **Table 2.4**.

D.2.9 Measuring the currents through the components and the phase angles with the oscilloscope

1. Using the same method you used in D.2.5 and D.2.6, use the given sense resistor to determine the peak values of the source current, the current through the resistor R1 and the current through the inductor L1. Record the values in **Table 2.4**.
2. For each case, note down the time (Delay) in **Table 2.4**.

**Fig.D.2.3: Parallel RLC circuit****D.2.10 Constructing Circuit 3 (Parallel RLC)**

1. Construct the circuit shown in **Fig.D.2.3** on the bread board. Use minimal wires.
2. Set 1 kHz Frequency and 3V peak Amplitude (6V peak to peak) in the Audio Generator.

D.2.11 Calculating the voltage across the circuit components

1. Calculate the magnitude and phase of I_S , I_C , I_L , and I_R and note down the values in **Table 2.6**.

D.2.12 Measuring the currents through the components and the phase angles with the oscilloscope

1. Using the same method you used in D.2.5 and D.2.6, use the given sense resistor to determine the peak values of the source current, the current through the resistor R1, the current through the capacitor C1 and the current through the inductor L1. Record the values in **Table 2.6**.
2. For each case, note down the time (Delay) in **Table 2.6**.

D.2.13 Comparing the practical and theoretical values of circuit currents

1. For each current value recorded so far, calculate the phase angle from the Delays (ΔT) using the given formula.
2. Calculate the difference between the practical and theoretical magnitudes and phase angles of the currents and record the values in the respective tables.

F.2 Simulation**F.2.1 Perform a Transient Analysis on the given parallel RLC circuit in Multisim**

1. Construct the circuit in Fig.D.2.3 in Multisim along with all the sense resistors you used.
2. Generate a set of graphs showing the time delays between the voltages across the different sense resistors and the source.
3. Now, remove the sense resistors and plot all the branch currents and the source current directly.
4. Attach the screen-shots of the circuit and the simulation results with your report.

G.2 Questions

1. A $1\text{k}\Omega$ sense resistor was used to perform this experiment. Suggest 1 possible advantage and 1 possible disadvantage of using an even smaller (say 10Ω) sense resistor in the first circuit (D.2.1).
2. Draw the phasor diagrams for the circuits in Fig D.2.1 and Fig D.2.2.
3. How would each of the phasor diagrams change if the source frequency was raised?
4. In case of the parallel RLC circuit, do the practical readings confirm the theoretical values? If any of the percentage differences are above 10%, suggest 3 possible reasons for the discrepancy.

E.2 Data Sheet: Lab 3, Experiment 2

Date:	Points:
Remarks:	

Signature of the Instructor

Student Information

Section:	Group:	Status:
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E.2.1 Table 2.1: Reactance and Impedance values (parallel RC circuit)

R (measured)(Ω)	C (measured)(F)	X_C (Theory) [$\frac{1}{2\pi fC}$] (Ω)	$ Z $ (Ω) [$\sqrt{R^2 + X^2}$]	$Z\angle\theta^\circ$ [$\tan^{-1}(\frac{X}{R})$]

E.2.2 Table 2.2: Comparing magnitudes and phases of i_C and i_R

	$ i_{\text{peak}} $ (Theory)	θ (Theory)	$V_{\text{sense(peak)}}$ (Measured)	$ i_{\text{peak}} $ (Practical)	Delay ΔT (Measured)	θ (Practical) [$\Delta T \times f \times 360$]	% Difference $ i $	% Difference θ
i_C								
i_R								
i_S								

E.2.3 Table 2.3: Reactance and Impedance values (parallel RL circuit)

R (measured)(Ω)	L (measured)(H)	X_L (Theory) [$2\pi fL$] (Ω)	$ Z $ (Ω) [$\sqrt{R^2 + X^2}$]	$Z\angle\theta^\circ$ [$\tan^{-1}(\frac{X}{R})$]

E.2.4 Table 2.4: Comparing magnitudes and phases of i_L and i_R

	$ i_{\text{peak}} $ (Theory)	θ (Theory)	$V_{\text{sense(peak)}}$ (Measured)	$ i_{\text{peak}} $ (Practical)	Delay ΔT (Measured)	θ (Practical) [$\Delta T \times f \times 360$]	% Difference $ i $	% Difference θ
i_L								
i_R								
i_S								

E.2.5 Table 2.5: Reactance and Impedance values (parallel RLC circuit)

R (Ω)	C (F)	L (H)	X_C (Theory) [$\frac{1}{2\pi fC}$] (Ω)	X_L (Theory) [$2\pi fL$] (Ω)	$ Z $ (Ω) [$\sqrt{R^2 + X^2}$]	$Z\angle\theta^\circ$ [$\tan^{-1}(\frac{X}{R})$]

E.2.6 Table 2.6: Comparing magnitudes and phases of V_C , V_L and V_R

	$ i_{\text{peak}} $ (Theory)	θ (Theory)	$V_{\text{sense(peak)}}$ (Measured)	$ i_{\text{peak}} $ (Practical)	Delay ΔT (Measured)	θ (Practical) [$\Delta T \times f \times 360$]	% Difference $ i $	% Difference θ
i_C								
i_L								
i_R								
i_S								